THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 2: **Linear Polynomials** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $p(x) = 7x + 3$ and $q(x) = -2x + 8$, then $p(x) + q(x) =$ 1

(a) $5x + 11$ **(b)** $5x + 12$ **(c)** $6x + 11$ **(d)** $5x + 10$

2. A linear polynomial $p(x) = ax + b$ has $p(1) = 5$ and $p(2) = 13$. Then $b =$ 1

(a) -4 **(b)** -1 **(c)** -2 **(d)** -3

3. The nth term of a pattern is $6n - 5$. Its 20th term is: 1

(a) 117 **(b)** 115 **(c)** 114 **(d)** 116

4. If $p(x) = ax + b$ and $b = 0$, the graph passes through: 1

(a) the x-axis only **(b)** $(1, 0)$

(c) $(0, 1)$ **(d)** the origin

5. If $x = 4$ is a zero of $8x + m$, then $m =$ 1

(a) -32 **(b)** -33 **(c)** -30 **(d)** -31

6. The slope of the line through $(2, 1)$ and $(6, 17)$ is: 1

(a) 5 **(b)** 3 **(c)** 6 **(d)** 4

7. The x-intercept of the line $y = 5x + 25$ is: 1

(a) -6 **(b)** -4 **(c)** -3 **(d)** -5

8. The point $(k, 13)$ lies on $y = 9x - 5$. Then $k =$ 1

(a) 2 **(b)** 4 **(c)** 1 **(d)** 3

9. If $p(x) = 3x + 9$ and $q(x) = 6x - 4$, then $p(x) + q(x) =$ 1

(a) $9x + 4$ **(b)** $9x + 5$ **(c)** $10x + 5$ **(d)** $9x + 6$

10. A linear polynomial $p(x) = ax + b$ has $p(1) = 17$ and $p(2) = 23$. Then $b =$ 1

(a) 13 **(b)** 12 **(c)** 11 **(d)** 10

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The nth term of a pattern is $9n - 4$. Find its first three terms and the 8th term. 2

12. Find the slope of the line through $(-2, -3)$ and $(2, 9)$. 2

13. For $p(x) = 9x + 2$ and $q(x) = 4x + 12$, find the zero of $p(x) - q(x)$. 2

14. The nth term of a pattern is $4n + 7$. Find its first three terms and the 25th term. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The line $y = mx + 24$ passes through $(1, 30)$. Find m, then the x-intercept of the line. 3

16. Find three points on the line $y = -6x + 7$ and show that the slope between the first and last is -6. 3

17. A pattern has first term 9 and common difference 8. Find its nth term and the term number whose value is 121. 3

OR

The line $y = mx + 21$ passes through $(2, 35)$. Find m, then the x-intercept of the line. 3

18. Find three points on the line $y = 10x - 9$ and show that the slope between the first and last is 10. 3

OR

A pattern has first term 20 and common difference 3. Find its nth term and the term number whose value is 107. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A linear polynomial $p(x)$ has $p(-1) = 8$ and $p(3) = 16$. Find $p(x)$, its zero and $p(7)$. 5

OR

A salesperson earns ₹3000 per month plus ₹80 for each item sold. Write her earning $E(n)$, find the earning for 30 items, the number of items sold for ₹9000, and the extra items needed to raise earning by ₹800. 5

20. A linear polynomial $p(x)$ has $p(4) = 7$ and $p(6) = 21$. Find $p(x)$, its zero and $p(2)$. 5

OR

A salesperson earns ₹8000 per month plus ₹300 for each item sold. Write her earning $E(n)$, find the earning for 10 items, the number of items sold for ₹20000, and the extra items needed to raise earning by ₹1500. 5

END OF QUESTION PAPER — Total: 40 Marks

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